

International Islamic University Chittagong

Morality Development Program (MDP)

Semester End Examination, Autumn- 2025

Course Title: Concepts of Moral Development

Course Code: MDP-3503, 5th Semester

Full Marks : 50

Time: 2.50 Hours

[Figures on the right hand margin indicate full marks]

Answer the following questions

Q.1

- a) On what context 'July Revolution 24' took place? 2
- b) Determine the contribution of Gen Z in 'July revolution 2024' 2
- c) Write a short note on the far reaching effect of 'July revolution 24' in the socio-political aspects of Bangladesh. 6

Q.2

- a) What is your understanding of an ideal society? 2
- b) On what ground students are considered as a very crucial community of a society? 3
- c) In what ways students can play a pivotal role in the formation of an ideal society? 5

Q.3

- a) Explain the Islamic & Western concepts of entertainment. 3
- b) What types of entertainments are allowed in Islam? 3
- c) Mention Shariah ruling on alcohol, pornography and free mixing. 4

Q.4

- a) Define 'Culture'. 2
- b) Figure out certain salient features of Islamic Culture and Western Culture? 4
- c) Mention some negative impacts of Western Culture in our everyday life. 4

or

Discuss the 7 categories of people who will get shelter under the shadow of Arsh on the Day of Judgment. 10

Q.5

- a) Explain Human rights in Islamic perspective. 2
- b) Describe the rights of minorities and other religious groups in an Islamic state. 4
- c) How to treat with the subordinates, orphans and destitutes in the light of Islam? 4

or

Explain the 5 questions that will be asked on the day of Resurrection. 10

International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination- Autumn -2025

Course Code: URED-3503(URED-2101 for LLB)
Course Title: Political thoughts and social Behavior

Full Marks: 50

Time: 2 Hours 30 Minutes

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	a. Explain the <u>objective</u> , <u>nature</u> , and <u>features</u> of marriage in Islam.	5		
	b. Summarize the essentials of a Muslim marriage.	5		
	Or,		3	Create
	c. Explain Quranic approach to matrimonial disputes.	5		
	d. "Divorce is last resort to dissolve a matrimonial disputes"- explain	5		
2	Evaluate the rights and duties of women in Islam. Critically asses how Islam seeks to maintain a balance between women's <u>right</u> and <u>responsibilities</u> in <u>spiritual</u> , <u>social</u> and <u>economic</u> domains.	10	⑧ 3	Evaluate
3	a. Describe how Islamic knowledge and its practices influence husband-wife relationship.	5		Evaluate
	b. Describe Islamic frameworks for parenting.	5		Remember
4	Critically evaluate the mechanism of Islamic economic system and assess how they promote justice and equity while preventing exploitation and excessive wealth concentration.	⑤ 10	3	Evaluate
5	a) "Only Allah (SWT) has the right to declare a thing lawful or prohibited"- verify.	5	3	Apply
	b) Illustrate the Islamic regulations of dress code for man and women.	5		Understand

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International Islamic University Chittagong
Department of Computer Science and Engineering
BSc in CSE Semester Final Examination, Autumn 2025
Course Code: CSE-3521 Course Title: Computer Architecture
Total Marks: 50 Time: 2 hours 30 Minutes

Figures in the right margin indicate full marks, course outcomes, and cognitive domains.

Group A		M	C	D
1.a)	Analyze the variation of Cycles Per Instruction (CPI) in single-cycle and multi-cycle processor architectures, and explain how instruction complexity influences CPI using a suitable instruction example.	5	3	4
OR				
1.a)	By describing the core operations of MIPS, draw the single cycle datapath for sub \$t3, \$s2, \$s3 explaining its data flow.	5	3	4
1.b)	Demonstrate the single-cycle datapath operations to execute a Load Word (LW) instruction, and illustrate the process using a suitable instruction-level example showing the required control actions.	5	3	2
OR				
1.b)	Explain the logic elements for multicycle datapath. Show and define how the elements handle edge-triggered clocking.	5	3	2
2.a)	Explain how the control unit generates signals for different instruction types in a multicycle-MIPS datapath.	5	3	2
2.b)	In a single-cycle MIPS datapath, a 16-bit immediate value is sign-extended to 32 bits before ALU operation. Given the immediate value: 1011 0000 0000 1101. Find the 32-bit sign-extended value and the decimal value of the sign-extended number.	5	2	3
Group-B				
3.a)	Given a sequence of dependent instructions, analyze how pipeline stalls are introduced and determine how reordering instructions can minimize performance degradation.	5	3	4
3.b)	Analyze how control signals are generated and propagated through the pipeline datapath, and explain how incorrect control handling can lead to hazards	5	3	4
4.a)	Describe the following with example: a) Multi-level Cache, b) 3Cs model.	5	2	2
4.b)	A cache consists of 64 blocks, each having a block size of 16 bytes. Determine the cache block number to which the byte address 1200 is mapped.	5	2	3
OR				
4.a)	A program frequently accesses a few recent variables and iterates over large arrays. Predict which parts benefit from temporal and spatial locality, and justify your reasoning.	5	2	2
4.b)	Explain how a direct-mapped cache uses memory block addresses, tag, bits, and valid bits to map a memory block to a cache block.	5	2	3
5.a)	Analyze how the high cost of a page fault influences the design decisions of virtual memory systems. How can you solve it?	5	2	4
5.b)	Analyze shortly on rotation latency, RAID, hot swapping, standby spares, DMA	5	2	4

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
Department of Computer Science & Engineering (CSE)
BSc in CSE, Final Examination, Autumn 2025

Course Code: **CSE-3527**
 Full Marks: **50**

Course Title: **Compiler**
 Time: **2 Hour 30 minutes**

[Answer all of the following questions from Group A and B.
 (Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

- | No. | Group – A [2×10 = 20 Marks] | M | COs | DL |
|-------|--|----------------|-----|-------|
| 1. a) | Syntax Analysis: Describe different type of syntax error that handling in C compiler construction and give the main error recovery strategies to handle these errors. | 2
(1+1) | CO3 | U |
| b) | Syntax Analysis: What is the impact of the left recursion for parsing in compiler construction? Is there any left recursion in the following grammar G_1 over the language $\Sigma = \{a, b, c, d\}$? If yes, eliminate the left recursion.
$A \rightarrow Ac \mid Aad \mid bd \mid \epsilon$ | 3
(1+2) | CO3 | An |
| c) | Consider the following grammar:
$S \rightarrow Bd \mid Cb$
$B \rightarrow aB \mid \epsilon$
$C \rightarrow cC \mid \epsilon$
i) Calculate the FIRST and FOLLOW sets for the following grammar
ii) Is the above grammar is LL(1) grammar? Justify your answer.
iii) Explain the significance of the "1" in LL(1). | 5 | CO3 | C |
| 2. a) | Consider the following Grammar:
$E \rightarrow E + T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$
Show the shift-reduce parsing steps for the input string: $id + id * id$ | 2 | | CO3 U |
| Or, | What are the conditions for a grammar to be an operator precedence grammar? | 2 | CO3 | U |
| b) | Syntax Analysis:
Consider the following grammar G_4 over the alphabet $\Sigma = \{a, b, c, d\}$. You want to use an CLR(1) parser for the strings defined by G_4 . You only need to provide accepting states, including their items, and transitions between accepting states. | 8
(3+2+1+2) | CO3 | C |

$$S \rightarrow AAa \quad A \rightarrow aA \mid bb$$

$I_0:$ $S' \rightarrow .S, \$$ $S \rightarrow .AAa, \$$ $A \rightarrow .aA, a/b$ $A \rightarrow .bb, a/b$
--

Fig. 2(b)

$I_0:$ $S' \rightarrow .S, \$$ $S \rightarrow .Aa, \$$ $A \rightarrow .bAc, \$$ $A \rightarrow .dc, \$$ $A \rightarrow .bda, \$$ $A \rightarrow .d, a$

Fig. 2(b) Or

- i) Provide the DFA transitions and state items of the LR(1) canonical collections. The first LR(1) DFA state item (I_0) already been provided in Fig. 2(b) for your reference.
- ii) Construct CLR(1) parser table. The parser table includes both the action and goto tables.
- iii) Is the grammar CLR(1)? Justify your answer.
- iv) Show the stack implementation along with the output parse tree for the input string 'abbbba'.
- Or,
- What do you mean by the conflict in a LR parsing table construction? Explain the possible types of conflict in a LR parsing and consider the following two grammar G_5 over the alphabet $\Sigma = \{a, b, c, d\}$.
- $$S \rightarrow Aa \mid bAc \mid dc \mid bda \quad A \rightarrow d$$
- Provide the DFA transitions and state items of the LR(1) canonical collections. The first LR(1) DFA state item (I_0) already been provided in Fig. 2(b) Or for your reference.
- i) Construct LALR(1) parser table. The parser table includes both the action and goto tables.
- ii) Show that above CFG G_5 is not SLR(1) grammar

Group – B [3×10 = 30 Marks]

3. a) Syntax directed translation: Write syntax-directed translation for the following grammar G_6 over the language $\Sigma = \{+, -, (,), id, num\}$ with every non-terminal has a synthesized attribute **.nptr** of pointer type for syntax tree constructions and write the basic functions to construct the syntax tree for arithmetic expression with example:

$E \rightarrow E + T \mid E - T \mid T \quad T \rightarrow (E) \mid id \mid num$

Give the syntax-directed definition of the following grammar G_5 . Construct the syntax tree with instructions to define pointer for the input expression string "a+5-c+1".

- b) Type checking:
- Write the basic examples of static checkers and show the placement of type-checker on compiler construction with a block diagram
 - What do you mean by "call by value" and "call by reference"
 - Write syntax-directed translation (semantic rules) for the following type checking statements syntax definitions:

$S \rightarrow \text{if } E \text{ then } S_1 \quad S \rightarrow \text{While } E \text{ do } S_1$

4. a) Run-time environment:
- What is the activation tree in the run-time environment of a compiler for a function?
 - What is the purpose of activation tree in the run-time environment of a compiler. Draw the activation tree for the following c programming function to generate the value of n^{th} factorial number, if $n=8$:

```
int factorial(int n){
    //Base case: if n is 0 or 1, return n
    if (n <= 1){
        return 1;
    }
    return n * factorial(n - 1);
}
```

- Explain the heap and stack allocation strategies for memory allocation

Or, What is the role of a **control stack**? Draw the activation tree for the following program:

```
main() {
    f();
}

f() {
    g();
    h();
}
```

- b) Intermediate code generation: Consider the following C program:

```
int main(){
    int a, b, c, d;
    scanf("%d%d%d", b, c, d);
    a = b * -c + b * -c + d; // '-' is unary minus
    print("%d", a);
    return 0; }
```

Translate the C assignment statement into:

- Syntax tree
- Three-address code
- Triples
- Quadruples
- DAG

Or, Translate the following expression to quadruple, triple, indirect triple and polish notation: $a + b * c / e + b * c$

5. a) Consider the following C program (Answer any one):

```
for (i=0; i<n; i++) {
    for (j=0; j<n; j++) {
        c[i][j] = 0.0; } }
for (i=0; i<n; i++) {
    for (j=0; j<n; j++) {
        for (k=0; k<n; k++) {
            c[i][j] = c[i][j] + a[i][k]*b[k][j]; } } }
```

Translate the above C program into three address code. Also construct basic block and flow graph for the above C code.

- b) Draw the syntax tree and DAG for the following expression: $a+a+(a+a+(a+a+a))$

International Islamic University Chittagong
Department of Computer Science and Engineering
BSc in CSE Semester Final Examination, Autumn 2025

Course Code: CSE-3523 Course Title: Microprocessors, Microcontrollers and Embedded Systems
Total Marks: 50 Time: 2 hours 30 Minutes

Figures in the right margin indicate full marks, course outcomes, and cognitive domains.

	0 1 2 3 4	Group A	M	C	D
1.a)		Discuss Type 3-5 interrupts of the 8086 microprocessor with an assembly code example for each.	4	2	N
1.b)		Explain the cause, operation, and execution steps of the IRET instruction in the 8086 ISR.	3	2	E
1.c)		Explain the Non-Maskable Interrupt (NMI) using a power failure detector as an example.	3	2	U
2.a)		Explain the role of a coprocessor, including its purpose and benefits, with a diagram.	5	2	N
2.b)		Explain arithmetic pipelining and instruction pipelining with examples.	5	2	N
		OR			
2.a)		What are pipeline hazards? Explain their main types and causes.	5	2	N
2.b)		Define pipelining and explain the stages of instruction pipelining with a diagram.	5	2	N
		Group B			
3.a)		Define memory expansion. Implement a $32K \times 16$ EPROM using $32K \times 8$ EPROMs, with a diagram.	3	3	E
3.b)		Explain the operating modes of the 82C55 PPI.	3	3	U
3.c)		Explain Mode 1 strobe input of the 82C55 PPI with figures.	4	3	N
4.a)		Describe the features and internal architecture of the 8051 microcontroller.	3	3	U
4.b)		Explain hardware-software co-design in embedded systems with a diagram.	3	3	N
4.c)		Design an FSM for a tea/coffee vending machine activated by a 5-taka coin, with options Tea, Coffee, or Cancel. Draw the state diagram.	4	3	E
		OR			
4.a)		Design an FSM for a vending machine serving tea, green tea, or coffee, including Cancel and coin return. Define all states and transitions.	3	3	U
4.b)		Using a home security system, explain co-specification and HW/SW partitioning with justification.	3	3	N
4.c)		Why can the 8051 access only 64 KB of ROM? Explain the role of the EA pin in ROM selection.	4	3	E
5.a)		Explain the core principles and characteristics of IoT with real-world relevance.	3	3	N
5.b)		Explain the core components of IoT with a layered architecture diagram.	3	3	E
5.c)		Explain M2M communication, its limitations, and how IoT overcomes them.	4	3	U

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Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2025
Course Code: EEE-2421 Course Title: Electrical Drives and Instrumentation
Total marks: 50 Time: 2 hours 30 minutes

[Answer **all** the questions. Figures in the right-hand margin indicate full marks.
 Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	BT	Marks
1.a	Explain the basic construction and working principle of an induction motor with proper net sketches.	1	U	5
1.b	A 4-pole, 3-phase induction motor operates from a supply with a frequency of 50 Hz. Calculate:	3	E	5
	i. The speed at which the magnetic field of the stator is rotating.			
	ii. The speed of the motor when the slip is 3%.			
	iii. The speed of the rotor when the slip is 0.04.			
	iv. The frequency of the rotor currents when the slip is 0.03.			
	v. The frequency of the rotor currents at standstill.			
OR				
1.a	When a 3- ϕ power supply is connected with the stator of a 3- ϕ Induction Motor then a rotating magnetic field creates and rotates around the stator. Explain how it happens? Also draw the transformer equivalent circuit of induction motor.	1	U	5
1.b	A 208-V, 10-hp, four-pole, 60-Hz, Y-connected induction motor has a full-load slip of 5 percent	3	E	5
	i) What is the synchronous speed of this motor?			
	ii) What is the rotor speed of this motor at the rated load?			
	iii) What is the rotor frequency of this motor at the rated load?			
	iv) What is the shaft torque of this motor at the rated load?			
2.a	Describe the operation of a current-source inverter (CSI) drive using the necessary block diagram.	CO2	R	5
2.b	Explain the full-step operation and half-step operation of a stepper motor with a suitable diagram.	CO3	C	5

Group B

Q. No.	Question	CLO	BT	Marks
3.a	Discuss the different types of errors and statistical analysis errors in a measurement system.	CO3	U	5
3.b	The expected value of the voltage across a resistor is 90 V. However, the measurement gives a value of 89 V. Calculate:	CO3	E	5
	a) Absolute error			
	b) Percentage error			
	c) Relative accuracy			
	d) Percentage of accuracy			
4.a	Analyze the application of the Instrumentation Amplifier in the Temperature Controller and Temperature Indicator with their circuit diagram.	CO2	An	5
4.b	Explain with the help of a functional block diagram, the operation of a digital voltmeter.	CO3	C	5

OR

4.a Describe the working principle of Digital Voltmeter (DVM) with its construction.

CO2

4.b Sketch common Cathode 7 segment display with appropriate truth table.

CO3

C

5.a Describe working principle of the Thermocouples and the Thermistor with proper net sketches.

CO3

U

5

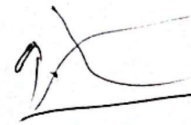
5.b Explain the operating principle of a photovoltaic cell or PV cell, using the necessary diagram.

CO2

An

5

$$R = R_0 (1 + \alpha \Delta T)$$



$T_{em} \uparrow$
 $R \downarrow$
e-free

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Autumn-2025, Final Examination

Course Code: CSE-3529, Title: System Analysis and Design

Full Marks: 30, Time: 2 Hour 30 Minutes

Answer all of the following questions

GROUP A

Marks

1. a) A College Cafeteria Management System is developed to manage food ordering and billing. (Students can view the food menu, place orders using their student ID) and collect food after payment. (The staff updates the menu and manages orders.)
- When an order is placed, the system records the order details and updates food availability. If an order is cancelled late, an extra charge is added. The system also prepares monthly sales reports for the cafeteria manager.
- i) Draw the Level 0 DFD (Context Diagram) of the system.
ii) Draw the Level 1 DFD, dividing the system into few sub-processes

5

CO1

OR of 1(a)

SmartHealth is one of the most widely used online healthcare management systems. The system has the following characteristics:

5

CO1

- Patients can register for a personal account by providing necessary personal information, username, and password to access healthcare services.
- Registered users can book doctor appointments, view medical reports, check appointment schedules, and manage their health records from their personal profiles.
- Doctors must log in to the system to view patient details, update diagnoses, and prescribe medications.
- All users are required to authenticate themselves through a secure login process before accessing the system.

Based on the above scenario, identify:

- i) Find all the external entities involved in the system.
ii) Design the Context Diagram or Level-0 Data Flow Diagram (DFD).
iii) Design the Level-1 Data Flow Diagram using proper balancing and leveling techniques.

- b) During Structured Analysis, various tools and techniques are used for system development. Discuss the following tools:

5

CO3

1. Data Flow Diagrams
2. Data Dictionary
3. Decision Trees
4. Decision Tables
5. Pseudocode

2. a) Explain the objectives of input design in a system. Differentiate between a well-designed input form and a poorly designed input form. Illustrate your answer using input form design examples.

5 CO3

b) What are the steps basics for designing output? Illustrate your answer using output design examples.

5 CO3

OR (of 2b)

An online property rental management system is widely used by property owners and tenants to manage rental listings and agreements. The system regularly updates property information and allows property owners to post rental properties with details such as rent, location, property type, and availability. Tenants can create personal accounts, sign in securely, search and filter properties by location, rent range, and property type, view property details, and submit rental applications online. The system also maintains lease records and payment history for approved rentals.

5 CO3

Draw an Entity Relationship Diagram (ERD) for the system based on the above scenario.

GROUP B

3. a) Explain the concept of system testing. Is it mandatory to perform system testing before deploying a software product? Support your answer with a real-life software example.

5 CO1

b) Define the following testing techniques and illustrate each with a suitable practical example:

5 CO3

1. Alpha Testing
2. Beta Testing
3. White Box Testing
4. Black Box Testing
5. Stress Testing

4. a) What is meant by system implementation? List and briefly explain the six key activities involved in ensuring a successful system implementation process.

5 CO1

b) As a systems analyst, you have been involved in a project to develop an inventory-tracking system for your business services office. The project is now coming to its final stages and you have been asked to write a training manual. Discuss main features of your training manual.

5 CO3

OR of 4(b)

EduCloud is an online learning management application that provides 24/7 access to courses, live classes, assignments, and student support for learners worldwide. The platform allows users to enroll in courses, submit assignments, attend virtual sessions, and make online payments for certifications.

5 CO3

Now, as a system analyst, what kinds of special considerations should be kept in mind for the maintenance of such an application?

5.	a)	What is threat? Explain about different kinds of threats. Write some criteria for software and hardware selection.				5	CO2																				
	b)	<table><tr><td></td><td>Database</td><td>Server</td><td>Application</td><td>Backup System</td></tr><tr><td>Admin</td><td>{read, write}</td><td>{execute}</td><td>{read, write, execute}</td><td>{read}</td></tr><tr><td>Manager</td><td>{read}</td><td>--</td><td>{read, write}</td><td>--</td></tr><tr><td>Employee</td><td>--</td><td>{execute}</td><td>{read}</td><td>--</td></tr></table>					Database	Server	Application	Backup System	Admin	{read, write}	{execute}	{read, write, execute}	{read}	Manager	{read}	--	{read, write}	--	Employee	--	{execute}	{read}	--	5	CO2
	Database	Server	Application	Backup System																							
Admin	{read, write}	{execute}	{read, write, execute}	{read}																							
Manager	{read}	--	{read, write}	--																							
Employee	--	{execute}	{read}	--																							
		You are given the above access control matrix. Show the Capabilities List and the Access Control List (ACL) derived from this matrix.																									
		OR of 5(b)																									
		Define system security. Discuss common security issues faced by computer systems with appropriate examples. Explain methods to prevent system vulnerabilities.				5	CO2																				